

DiffServ-Based QoS Solutions for Enhancing VoIP Performance in Enterprise Networks

Traffic Prioritization and Codec Analysis for Optimized Communication

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Abstract—This paper investigates the implementation of Differentiated Services (DiffServ)-based Quality of Service (QoS) mechanisms to enhance the performance of Voice over Internet Protocol (VoIP) in enterprise networks. VoIP traffic is highly sensitive to latency, jitter, and packet loss, making QoS essential for maintaining high-quality communication. The study is conducted in three phases: setting up a basic VoIP-enabled LAN, extending the network for traffic monitoring and Best Effort (BE) analysis, and implementing a DiffServ domain to prioritize VoIP traffic. Key components include traffic classification using Access Control Lists (ACLs), policy-based bandwidth allocation, and codec analysis under varying network conditions. Results demonstrate that DiffServ effectively prioritizes VoIP traffic, ensuring consistent throughput and superior Quality of Experience (QoE) even under congested scenarios. Codec performance is analyzed, highlighting the trade-offs between bandwidth requirements and audio quality. The findings underscore the importance of QoS in modern enterprise networks, providing a scalable solution to address network congestion and optimize VoIP performance. Figures, equations, and experimental data further support the conclusions drawn.

Keywords—VoIP, Quality of Service (QoS), Differentiated Services (DiffServ), Traffic Prioritization, Best Effort (BE), Codec Analysis, Enterprise Networks, Network Congestion, Quality of Experience (QoE), Bandwidth Allocation

I. INTRODUCTION

Voice over Internet Protocol (VoIP) has revolutionized communication by enabling voice calls over IP networks, making them more cost-effective and scalable compared to traditional telephone systems. However, the performance of VoIP services depends heavily on the quality of the underlying network. Issues like packet loss, delay, jitter, and bandwidth constraints can significantly degrade call quality, impacting the overall user experience. To address these challenges, Quality of Service (QoS) techniques are essential for prioritizing VoIP traffic and ensuring a reliable communication experience.

This study focuses on designing, implementing, and optimizing a VoIP-enabled enterprise network using QoS mechanisms to improve the Quality of Experience (QoE). The project was divided into three distinct but interconnected tasks,

each addressing a specific aspect of network configuration and performance enhancement:

- Task 1: A basic Local Area Network (LAN) was designed and set up to support VoIP services. Devices were configured with static IPs, and an Asterisk server was implemented for managing VoIP traffic. The task ensured connectivity and tested VoIP functionality within the local network.
- Task 2: The network was extended by adding additional subnets to simulate a more realistic enterprise environment. This stage involved analyzing the effects of Best Effort (BE) traffic on VoIP performance, exposing the limitations of networks without prioritization mechanisms.
- Task 3: Differentiated Services (DiffServ) was implemented to introduce QoS policies for prioritizing VoIP traffic. Traffic was classified and managed using policy-based routing and bandwidth allocation, ensuring that latency-sensitive VoIP traffic received the necessary priority over other types of data.

In addition, the study analyzed the impact of different codecs, such as PCMA, G.722, and iLBC, on VoIP performance under various bandwidth constraints. Tools like Wireshark and iPerf were used to capture traffic data and generate network loads, providing insights into throughput, delay, and packet loss for VoIP and other services.

By examining the role of QoS in mitigating the challenges faced by VoIP services, this project demonstrates how DiffServ-based traffic prioritization can ensure high call quality even in congested networks. The findings are particularly relevant for enterprise networks where maintaining reliable communication is critical. This paper serves as a practical guide for network engineers aiming to optimize VoIP performance in real-world deployments..

II. OBJECTIVES

The primary objective of this project is to design, implement, and optimize a VoIP-enabled enterprise Local Area Network (LAN) using Quality of Service (QoS) mechanisms to enhance the Quality of Experience (QoE) for VoIP services. The study

aims to address common networking challenges, such as packet loss, jitter, delay, and bandwidth constraints, which can degrade VoIP performance.

A. Establish a VoIP-Enabled LAN (Task 1)

This task involves designing and configuring a basic LAN topology with NAT for internet access. An Asterisk server is implemented for VoIP services, enabling connectivity among devices like VoIP phones and softphones. The successful deployment of VoIP calls and network connectivity is verified through testing.

B. Analyze Best Effort (BE) Traffic Behavior (Task 2)

The network is extended to include multiple subnets, creating a realistic enterprise topology. BE traffic is simulated using iPerf to generate network congestion, and its impact on VoIP performance is analyzed. Different VoIP codecs, including PCMA, G.722, and iLBC, are evaluated under varying bandwidth conditions to assess their performance.

C. Implement QoS with Differentiated Services (DiffServ) (Task 3)

DiffServ is configured to classify and prioritize VoIP traffic using Access Control Lists (ACLs) and policy maps. Bandwidth is allocated to ensure consistent throughput and minimal latency for VoIP traffic under heavy load conditions. The effectiveness of QoS mechanisms is analyzed using tools like Wireshark to monitor traffic flow and measure network performance.

Performance Analysis and Comparison. VoIP performance is compared in BE and QoS-enabled scenarios. The study assesses QoE for VoIP services under different configurations, highlighting trade-offs between bandwidth usage and voice quality for selected codecs. By systematically addressing these objectives, the study provides a comprehensive understanding of how QoS mechanisms can enhance VoIP performance in enterprise networks.

III. METHODOOGY

The methodology for this study involved designing, implementing, and testing a VoIP-enabled enterprise Local Area Network (LAN) with Quality of Service (QoS) mechanisms to optimize VoIP performance. The project was divided into three distinct tasks, each focusing on specific aspects of network configuration and analysis. The process was executed systematically to ensure comprehensive evaluation and validation of the network's functionality.

LAN A IP address: 10.9.0.0 /24

Below are the IP addresses of devices in LAN A

Table I.

LAN A	Ip Addresses of device		
		Subnet Address	Subnet Mask
Default Gateway	1st available IP address	10.9.0.1	
Asterisk / Iperf Server	2nd available IP address	10.9.0.2	255.255.255.0
VoIP Phone	3rd available IP address	10.9.0.3	255.255.255.0

LAN A	Ip Addresses of device		
		Subnet Address	Subnet Mask
PC Softphone	4th available IP address	10.9.0.4	255.255.255.0
Outside interface (g0/0/0)		139.6.19.50	

A. Task 1: Establishing a Basic LAN with Internet Access and VoIP Service

- **LAN A Subnetting:** The initial phase involved creating a functional LAN (LAN A) with internet connectivity and VoIP capabilities. The network was designed using a private IP address space (10.9.0.0/24) and included devices such as a router, switch, Asterisk server, VoIP phone, and a PC running softphone software.

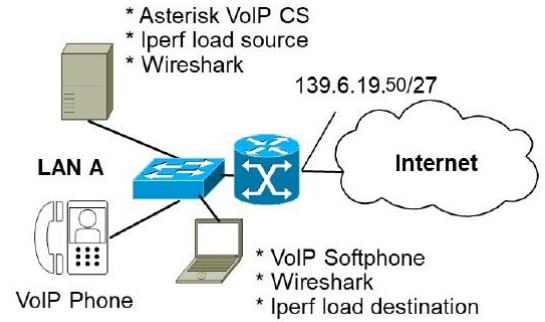


Fig 1: Basic LAN with Internet Access and VoIP Service.

- **Network Topology and Initialization:** The LAN A topology included a router, switch, Asterisk server, and VoIP phone. Devices were interconnected via a patch panel, with the switch's G0/1 interface connected to the router's G0/0/1 (default gateway) and G0/0/0 linked to an uplink switch for Internet access. Static IP addresses, subnet masks, and default gateway settings were assigned, and connectivity was verified through ping tests.
- **Switch Configuration:** The switch was accessed via a console cable. Configurations included assigning the hostname "S1," disabling DNS lookups, setting local passwords, and adding a message of the day (MOTD) for security. The configuration was saved to non-volatile random access memory (NVRAM).
- **Router Configuration:** The router was configured with the hostname "R1," Telnet passwords, and encrypted credentials. Interface descriptions and IP addresses were set, and the configuration was saved to NVRAM. Connectivity was confirmed through testing.
- **Network Address Translation (NAT):** NAT was configured to enable Internet access for LAN A. An ACL defined the private IP range, NAT inside and outside interfaces were set, and a static default route was configured. Internet connectivity was verified through ping tests.
- **VoIP Service Implementation:** VoIP services were implemented using an Asterisk server. Local users and call

routing were configured, and hard phones and softphones were set up with appropriate network credentials. VoIP

The network demonstrated full connectivity within LAN A, seamless Internet access via NAT, and proper operation of VoIP services with successful call tests. The project successfully established a QoS, achieving all objectives through effective network configuration and NAT implementation.

B. Task 2: Extending the Network and Analyzing Best Effort Traffic

In the second phase, the network was expanded to simulate a larger enterprise environment by adding two additional subnets, LAN B (10.9.1.0/24) and LAN C (10.9.2.0/24), connected via Router R1 and Router R2. A low-bandwidth serial link (128 kbps) connected LAN B and LAN C to replicate bandwidth constraints

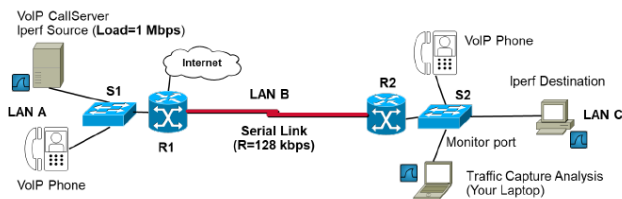


Fig 2: Extended Network for Analyzing Best Effort Traffic.

- **Extend and cable the network:** The network is extended by Connecting S2 with Fast Ethernet port Fa0/1 to a Gigabit Ethernet port of R2. The cabling of one VoIP Phone and the Iperf traffic load destination PC is moved to switch S2 in LAN C. The VoIP Phone is connected to S2 port Fa0/2 and the traffic load destination PC to S2 port Fa0/3. A serial link between routers R1 and R2 is established with V.35 cables.
- **LAN B and LAN C Subnet Addressing:** LAN B has been assigned the private IP address space of 10.9.1.0/24 and LAN C has been assigned the private IP address space of 10.9.2.0/24. R1 serial Interface is assigned IP address 10.9.1.1 and subnet mask of 255.255.255.0 and R2 serial interface is assigned IP address 10.9.1.2 and subnet mask of 255.255.255.0. Devices in LAN C has been assigned following IP addresses 10.9.2.1 for default gateway, 10.9.2.2 for VoIP phone, 10.9.2.3 for Iperf load destination and 10.9.2.4 for PC soft phone.
- **Configure LAN B and LAN C:** A low bandwidth serial link is implemented between LANs to emulate low bandwidth internet connections. A serial connection requires a clock signal on the line. The clock master is generated in the DCE interface. The clock slave interface is called DTE. The R1 serial interface is master DCE. The clock rate command for 128 kbps link bandwidth, only required for DCE interface. R2 serial interface is DTE interface. The routers' interfaces were configured with IP addresses and subnet masks appropriate to each LAN. Static routes were implemented. Router R2: Default route to unknown networks via R1. Router R1: Route to LAN C

using the active serial interface. Routing tables were verified to ensure connectivity across subnets

- **Traffic Monitoring:** A mirror port is configured on switch S2 for monitoring network traffic. A mirror port in a Switch is used to copy all Ethernet frames, send and received, from selected switch ports to a separate switch port, which is connected with a monitoring device. In our case this is the Monitor-PC. The mirror port cannot be used for normal connectivity, but will only be available for monitoring purposes.
- **Analyze best effort (BE) behavior for VoIP / iPerf traffic:** Three VoIP codecs -PCMA, G.722 and iLBC, which are supported by VoIP phones were selected to evaluate application and data link layer throughputs: Application throughput was measured for each codec. Data link throughput was analyzed using Wireshark captures. Because the VoIP service uses UDP encapsulation, we generated UDP traffic load of 1 Mbps for duration of 10ms in parallel to check the bandwidth sharing behavior in a best effort (BE) scenario on a 128-kbps serial link.

C. Task 3: Implementing QoS with DiffServ

In this part a details step-by-step approach to implementing and analyzing QoS for VoIP services in a structured enterprise network. Tasks included topology design, ACL configurations, and traffic scheduling, with a focus on ensuring expedited forwarding (EF) for VoIP traffic and assured forwarding (AF) for best-effort services

- **Service Definition:** The network runs 2 network services
 - VoIP (or voice and video): including signaling traffic and media data (sample) traffic. The VoIP service uses a PCM codec for sampling and SIP signaling.
 - Best effort service for any other traffic: Both routers are ingress routers in the DiffServ Domain. Monitoring is configured in LAN C. Thus any traffic from LAN A to LAN C is considered for analysis. SIP signaling and RTP media data are sent and received to and from the Asterisk server. Iperf load is used to check the QoS behavior and traffic prioritization in the DiffServ Domain.
 - **DiffServ Domain Implementation**
- To ensure effective traffic prioritization, the DiffServ architecture was employed. The VoIP media traffic was assigned the Expedited Forwarding (EF) class with a Differentiated Services Code Point (DSCP) value of 46, while signaling traffic was categorized under Assured Forwarding (AF12) with a DSCP value of 12. All remaining traffic was classified as Best Effort (BE) with a DSCP value of 0.

Table II.

Service	Class Name	DSCP Class	DSCP Value
---------	------------	------------	------------

VoIP - Media	Expedited Forwarding	EF	46
VoIP - Signaling	Assured Forwarding	AF12	12
Best Effort	Default Forwarding	BE	0

• Qos Scheduling and Bandwidth Allocation

VoIP traffic is guaranteed link layer throughput with 10% additional bandwidth reserved for signaling.

- VoIP RTP bandwidth: 85.6 kbps.
- Signaling bandwidth: 8.56 kbps
- Total: 94.16 kbps.

The VoIP service should have minimum latency and guaranteed bandwidth for one VoIP call, when VoIP traffic is forwarded on a router link. Policy map is created for R1 and R2 egress interfaces for bandwidth allocation and to give priority to VoIP traffic.

• QoS Implementation

Access Control Lists (ACLs) were configured on both routers (R1 and R2) to classify VoIP and Iperf traffic, enabling accurate mapping of flows to DiffServ classes. Class maps and policy maps were subsequently defined to assign QoS attributes and enforce traffic treatment policies.

Traffic prioritization was achieved through policy maps applied to ingress and egress interfaces. Ingress policies ensured the correct DSCP values were assigned based on traffic classification, while egress policies allocated bandwidth and provided guaranteed throughput for VoIP traffic. Priority was given to RTP media traffic, with additional bandwidth reserved for SIP signaling.

IV. RESULTS AND DISCUSSION

1. Analysis of Wireshark Capture during VoIP call:

The Wireshark capture showcases a VoIP call using the PCMA (G.711) codec between two phones ('10.9.0.2' and '10.9.0.3').

Fig 3: Sip establishment and Rtp packet Demonstration.

- SIP Signaling Traffic:** SIP messages ('INVITE', '100 Trying', '180 Ringing', '200 OK', 'ACK') indicate smooth call establishment without retransmissions. Minimal signaling latency suggests stable communication.
- RTP Traffic: Codec:** PCMA (G.711) with an application layer throughput of 64 kbps and data link throughput of 85.6 kbps. Steady RTP packet flow with no packet loss or jitter observed, confirming high-quality media transmission.

The PCMA codec uses significant bandwidth, with consistent throughput at 85.6 kbps. Network conditions are optimal, with no packet loss or jitter, ensuring excellent call quality

2. Analysis of Best Effort (BE) Behavior for VoIP and iPerf Traffic

The analysis of Best Effort (BE) behavior was conducted using three VoIP codecs—PCMA, G.722, and iLBC—under a simulated BE scenario on a constrained 128 kbps serial link. The objective was to evaluate the application and data link layer throughput for each codec and analyze the bandwidth-sharing behavior with a 1 Mbps UDP traffic load generated in parallel.

Table III.

Codec	Sampling rate	Application Layer Throughput in kbps	Data Link Layer Throughput in kbps
PCMA	8000	64	85.6
G.722	16000	64	85.6
iLBC	8000	15.2	27.4

Throughput Calculation

• Overhead Calculation:

$$\text{RTP} + \text{UDP} + \text{IP} + \text{Ethernet} = 12 + 8 + 20 + 18 = 58 \text{ Bytes}$$

• Total Packet Size:

$$\text{Payload} + \text{Overhead} = 160 + 58 = 218 \text{ Bytes}$$

• Packet Rate:

$$\text{Packets/second} = \frac{1}{20 \text{ ms}} = 50$$

• Throughput:

$$50 \times 218 \times 8 = 87.2 \text{ kbps}$$

Thus, the calculated data link layer throughput is 87.2 kbps

Key Observations

• Application Throughput for Each Codec:

At the application layer, both PCMA and G.722 achieved consistent throughput of 64 kbps due to their high sampling rates and encoding characteristics. The iLBC codec, optimized for low-bitrate scenarios, delivered a significantly lower application throughput of 15.2 kbps.

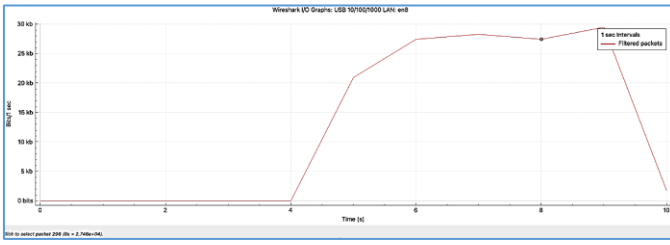


Fig 4: I/O graph of iLBC code.

• Data Link Layer Throughput (Wireshark Analysis):

Wireshark captures revealed that PCMA and G.722 codecs had a data link throughput of 85.6 kbps, which includes additional overhead from UDP encapsulation and network layer protocols. The iLBC codec had a lower data link throughput of 27.4 kbps, reflecting its smaller payload size and reduced overhead.

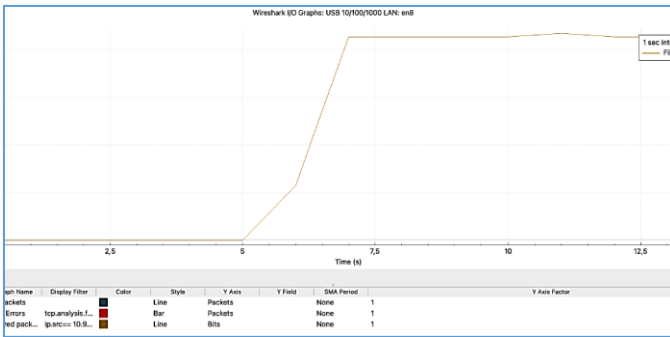


Fig 5: I/O graph of G711 code.

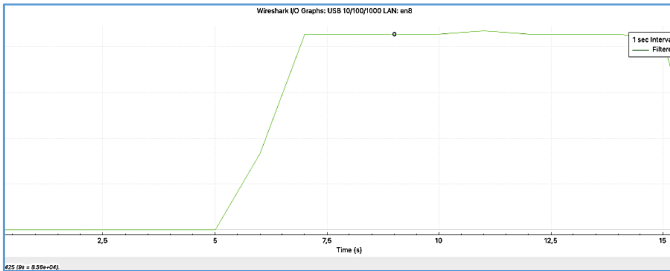


Fig 6: I/O graph of PCMA code.

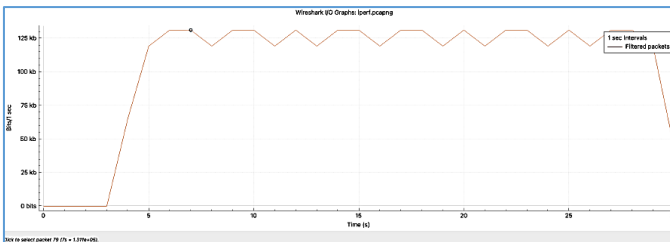


Fig 7: I/O graph of iPerf.

• 3. Bandwidth Sharing in BE Traffic:

The introduction of a 1 Mbps UDP traffic load in parallel demonstrated significant competition for bandwidth. Wireshark analysis showed that:

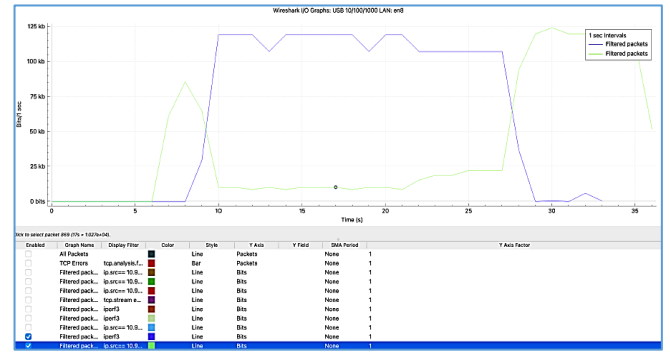


Fig 8: I/O graph of iPerf with VoIP(PCMA)

The 128 kbps serial link reached saturation during testing, with BE traffic consuming the majority of the available bandwidth. This caused increased jitter, packet loss, and reduced throughput for PCMA and G.722 codecs, severely impacting VoIP Quality of Experience (QoE).

- PCMA and G.722 experienced noticeable degradation in throughput under BE conditions due to their higher bandwidth requirements.
- iLBC showed better resilience under congestion, maintaining a lower but more stable throughput.

3. Bandwidth Allocation in Best Effort (BE) Scenario for G.722 Codec

For a 128 kbps serial link, traffic is shared proportionally based on load:

• Total Bandwidth:

$$\text{Total} = 119.2 \text{ kbps (iPerf)} + 10.27 \text{ kbps (VoIP)} = 129.47 \text{ kbps}$$

• Proportional Ratios:

$$\text{iPerf: } \frac{119.2}{129.47} = 92.1\%, \quad \text{VoIP: } \frac{10.27}{129.47} = 7.9\%$$

• Allocated Bandwidth:

$$\text{iPerf: } 92.1\% \times 128 = 117.89 \text{ kbps,}$$

$$\text{VoIP: } 7.9\% \times 128 = 10.11 \text{ kbps}$$

Results:

Table IV.

Link Bandwidth	Iperf Traffic Data Link Layer Throughput in kbps	VoIP Traffic Data Link Layer Throughput in kbps (PCMA)
128 kbps	117.89kbps	10.11kbps

The results show VoIP traffic receives a slightly higher share compared to other scenarios but remains constrained, emphasizing BE's limitation for real-time communication.

The results clearly demonstrate the limitations of Best Effort traffic handling for VoIP services on constrained links. The severe degradation observed for PCMA and G.722 underscores the necessity of traffic prioritization mechanisms such as DiffServ. iLBC, while more resilient, sacrifices audio quality, making it suitable only for environments where low bandwidth is unavoidable.

4. DiffServ Domain Implementation and QoS Traffic Analysis:

The implementation of QoS using DiffServ on routers R1 and R2 significantly improved VoIP performance under high network load. By prioritizing VoIP traffic over iPerf load traffic, consistent throughput and low latency were ensured.

- **VoIP Bandwidth Allocation:**

RTP Bandwidth:

85.6 kbps

Signaling Bandwidth:

$10\% \times 85.6 = 8.56 \text{ kbps}$

Total VoIP Bandwidth:

$85.6 + 8.56 = \mathbf{94.16 \text{ kbps}}$

Remaining Bandwidth for iPerf Traffic:

$\text{iPerf Bandwidth} = 128 - 94.16 = \mathbf{33.84 \text{ kbps}}$

These results confirm that QoS prioritization ensures sufficient resources for VoIP traffic while effectively managing iPerf traffic within the remaining bandwidth. Policy maps allocated this bandwidth to VoIP traffic, ensuring its prioritization with Expedited Forwarding (EF), while iPerf traffic was assigned Default Forwarding (DF)

- **Wireshark Observations:**

- VoIP traffic maintained a steady throughput of 85.6 kbps, fully utilizing its allocated bandwidth.

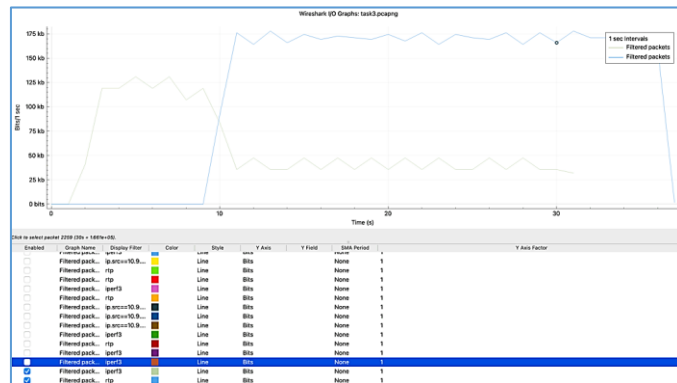


Fig 9: I/O graph of iPerf.

iPerf traffic was limited to 41.72 kbps, confirming its lower priority. The Wireshark I/O graph illustrates this prioritization, with VoIP traffic throughput remaining unaffected by competing load traffic.

- **Quality of Experience (QoE):**

The VoIP call using the PCMA codec achieved excellent clarity, minimal delay, and no echo, earning a QoE rating of Excellent. The results highlight the effectiveness of DiffServ in ensuring reliable VoIP communication in enterprise networks. QoS prioritization allowed VoIP traffic to maintain high-quality performance even under heavy load, as evidenced by the Wireshark analysis. These findings underscore the importance of QoS mechanisms in addressing the challenges of managing latency-sensitive applications.

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CONCLUSION

This study demonstrated the importance of QoS in ensuring reliable VoIP performance in enterprise networks. Task 1 established a functional VoIP-enabled LAN, Task 2 highlighted the impact of Best Effort traffic on VoIP quality, and Task 3 implemented DiffServ to prioritize VoIP traffic. The results confirmed that QoS effectively mitigates packet loss, jitter, and latency, maintaining excellent call quality even under heavy network load. These findings emphasize the necessity of traffic prioritization for real-time applications in mixed traffic environments.

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Task 1: Command Reference**Switch Configuration**

```
>hostname S1
>no ip domain-lookup
>enable secret class
>line con 0
>password cisco
>login
>banner motd # Unauthorized access
prohibited #
>copy running-config startup-config
```

Router Configuration

```
>hostname R1
>enable secret class
>line vty 0 4
>password cisco
>login
>service password-encryption
>int g0/0/0
>description Connection to Internet
ip address <public IP> <subnet mask>
>no shut 10.9.0.1 255.255.255.0
>int g0/0/1
>description Connection to LAN A
>ip address <LAN gateway IP> <subnet
mask>
>no shut
>copy running-config startup-config
```

NAT Configuration

```
>access-list 1 permit 10.<team-no.>.0.0
0.0.255.255
>ip nat inside source list 1 interface
g0/0/0 overload
>int g0/0/0
>ip nat outside
>int g0/0/1
>ip nat inside
>ip route 0.0.0.0 0.0.0.0 g0/0/0
```

VoIP Service Configuration**SIP Configuration (sip.conf)**

```
[general]
context=default
allowguest=no
srvlookup=yes
videosupport=yes
```

6001

```
[4001]
type=friend
host=dynamic
secret=4001 6001
context=internal
callerid="4001" <4001> 6001
allow=g722
directmedia=yes
```

```
[4002]
type=friend
host=dynamic
secret=4002 6002
context=internal
callerid="4002" <4002> 6002
allow=g722
video=all
directmedia=yes
```

Extensions Configuration (extensions.conf)

```
[general]
static=yes
writeprotect=no
```

```
[internal]
exten => 6001,1,Dial(SIP/6001,20)
exten => 6002,1,Dial(SIP/6002,20)
```

```
; Optional: Add hangup handling
exten => h,1,Hangup()
```

```
; Video call handling - ensure Asterisk
negotiates video
exten => 6001,2,Set(VIDEO_CALL=1)
exten => 6002,2,Set(VIDEO_CALL=1)
```

The SIP and extension configurations for local users, supporting video capabilities and defining proper call routing.

Task 2: Command Reference**Router R1 Serial Interface Configuration**

```
>int s0/x/0 s0/1/0
>description LAN B Interface
>ip address <your IP address> <your mask>
>clock rate 125000 10.9.1.1 255.255.255.0
>no shut
>exit
>ip nat inside
>copy running-config startup-config
```

Router R2 Serial Interface Configuration

```
>int s0/x/0    s0/1/0
>description LAN B Interface
>ip address <your IP address> <your mask>
>no shut      10.9.1.2 255.255.255.0
>exit
>copy running-config startup-config
```

Router R2 Gigabit Ethernet Interface Configuration

```
>int g0/0/0    g0/1
>description LAN C Interface
>ip address <your IP address> <your mask>
>no shut      10.9.2.1 255.255.255.0
>exit
>ip route 0.0.0.0 0.0.0.0 s0/1/0
```

Static Route in R1

```
>ip route 10.9.2.0 255.255.255.0 s0/1/0
```

Mirror Port Configuration on S2

```
>monitor session 1 source interface G0/0
>monitor session 1 destination interface
>Fa0/24
>copy running-config startup-config
```

Traffic Load Generation

```
iperf3 -c 10.9.2.3 -u -b 1M -t 10
```

Task 3: Command Reference

Router R1 Configuration

Access Control Lists (ACLs):

```
>access-list 101 permit udp 10.9.0.2
0.0.0.255 eq 5060 10.9.2.2 0.0.0.255 eq
5060
>access-list 102 permit udp 10.9.0.0
0.0.0.255 10.9.2.0 0.0.0.255 eq 50040
>access-list 103 permit ip any any
```

Class Maps:

```
class-map match-all rtp
  match access-group 102
class-map match-all sip
  match access-group 101
class-map match-all BE
```

```
match access-group 103
```

Policy Maps:

Ingress:

```
policy-map RlingressSETDSCP
  class rtp
    set ip dscp 46
  class sip
    set ip dscp 12
  class BE
    set ip dscp 0
interface GigabitEthernet0/1
  service-policy input RlingressSETDSCP
```

Egress:

```
policy-map Rlegress
  class rtp
    priority 90
  class sip
    bandwidth percent 10
interface Serial0/1/0
  service-policy output Rlegress
```

Router R2 Configuration

Access Control Lists (ACLs):

```
>access-list 104 permit udp 10.9.2.2
0.0.0.255 eq 5060 10.9.0.2 0.0.0.255 eq
5060
>access-list 105 permit udp 10.9.2.2
0.0.0.255 eq 50040 10.9.0.3 0.0.0.255
>access-list 106 permit ip any any
```

Class Maps:

```
class-map match-all rtp
  match access-group 105
class-map match-all sip
  match access-group 104
class-map match-all BE
  match access-group 106
```

Policy Maps:

Ingress:

```
policy-map R2ingressSETDSCP
  class rtp
    set ip dscp 46
  class sip
    set ip dscp 12
```



```
class BE
  set ip dscp 0
interface GigabitEthernet0/1
  service-policy input
  R2ingressSETDSCP

Egress:

policy-map R2egress
  class rtp
    priority 90
  class sip
    bandwidth percent 10
interface Serial0/1/0
  service-policy output R2egress
```

iLBC		1.10	Voice echoes. Speech quality is clear and audible.	4th
		1.80		7/10
	128	0.90	There is little echo in voice.	3rd
		0.71		8/10
		0.75	Speech quality is audible, could be better.	

Perceptual impression and rating based on Quality-of-Experience (QoE) For each link bandwidth and codec:

Codec	Link Band-width in kbps	QoE Description		Rate and Ranking
		Delay (seconds)		
PCMA	32	4.41	Voice echoes. Speech quality is very poor. Takes time to make call.	6th
		5.20		4/10
		9.7		
	64	4.19	Voice echoes. Speech quality better than the previous.	5th
		4.91		5/10
		5.92		
	128	0.75	There is little echo in voice.	1st
		0.95		10/10
		0.82	Speech quality best, loud, clear and precise.	
	32	0.52	Voice echoes. Speech quality is clear.	2nd
		0.45		9/10
		1.10		
	64	1.26		

